

Nikhil Kumar Reddy Pyreddy

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SUMMARY

Full-stack software engineer with five years of experience building production web applications and AI-integrated backend services. Currently developing LLM-powered search and AI agent tooling at Paycom, including building Server-Sent Events (SSE) streaming layers for real-time AI responses, integrating with locally hosted models, and consuming FastAPI ML service endpoints. Experienced across C#/.NET, Python, TypeScript, and React, with production deployments on Kafka, Kubernetes, and AWS. Strong background in API design, developer tooling, and shipping features end-to-end in Agile/SAFe environments.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), C#, Java, SQL

Frontend: React (Redux Toolkit, Hooks, Context API), Angular, HTML5, CSS3, Bootstrap

Backend & APIs: FastAPI, .NET 8, ASP.NET Core (Web API, MVC), Spring Boot, Node.js, Flask, REST, GraphQL

AI & ML Integration: LLM APIs, LangChain, RAG pipelines, prompt engineering, MCP integrations, GitHub Copilot

Cloud & Infrastructure: AWS (Lambda, S3, CloudWatch), Docker, Kubernetes, HashiCorp Vault, CI/CD (GitHub Actions, Azure DevOps)

Data & Streaming: Apache Kafka, Server-Sent Events (SSE), MS SQL Server (T-SQL), MySQL, PostgreSQL, MongoDB, Redis

Testing & Observability: xUnit, Moq, JUnit, Jest, React Testing Library, SonarQube, NLog, Splunk

PROFESSIONAL EXPERIENCE

Software Developer | *Paycom Software, Inc.* | Grapevine, TX

Jun 2024 to Present

- Built the AI-powered search integration for Paycom's Knowledge Base Help Center by implementing a Server-Sent Events streaming layer that delivers real-time, token-by-token LLM responses to a React and TypeScript frontend with backpressure handling, connection recovery, and graceful degradation when model latency spikes.
- Integrated with locally hosted Kimi and Qwen models deployed on-premises as AI agent tools by building the service layer that routes queries through RAG pipelines, applies prompt templates, and streams structured responses back to the user.
- Consumed and extended FastAPI endpoints exposed by the ML services team, building adapter layers in C#/.NET that bridge internal LLM inference services with the Knowledge Base product's search and content discovery workflows.
- Designed and built Kafka producer and consumer services that sync Knowledge Base content with the LLM services for AI-powered search and extended the integration with MCP endpoints exposing internal KB content and tooling to AI agents.
- Developed and maintained backend services and frontend features for the Knowledge Base Help Center and Advanced Report Writer products using C#/.NET 8, React, and TypeScript, including reusable UI components, typed API integrations, and state-managed user workflows, serving Paycom's 36,000+ enterprise clients.
- Designed the data-access layer using Dapper with a shared wrapper and AutoMapper for entity-to-DTO mapping, then standardized these patterns into reusable internal libraries adopted by other feature teams.
- Led the Confluence-to-Payges Knowledge Base platform migration covering 6,000+ articles by building API integration adapters and validation utilities, with the tooling automatically catching 97% of migration issues before manual review.
- Deployed services to Kubernetes using Docker containerized builds with secrets managed in HashiCorp Vault, CI/CD pipelines through GitHub Actions with SonarQube quality gates, and AWS Lambda for serverless workflows.

Software Developer - Graduate Research | *SLB - The University of Oklahoma* | Norman, OK

Aug 2022 to May 2024

- Owned end-to-end development of two industry-sponsored engineering tools for SLB Cameron's mechanical equipment teams, leading the work from requirements gathering through architecture, implementation, and stakeholder validation.
- Designed and built a parametric simulation tool from scratch in Python with 50+ configurable input variables, defining the data model, validation logic, and a CLI interface for batch execution by domain engineers.
- Built ETL and data visualization pipelines using Python, NumPy, Pandas, and Matplotlib to replace manual spreadsheet workflows with automated, repeatable analysis that reduced processing time from hours to minutes.
- Implemented multi-threaded batch processing for large parameter sweeps using Python's concurrent.futures, achieving 4x throughput improvement over sequential execution.

Software Development Intern | *Paycom Software, Inc.* | Grapevine, TX

May 2023 to Aug 2023

- Developed customer data import workflows in C# using ASP.NET Core MVC with REST API integration, SQL Server stored procedures, and unit and integration testing, reducing data quality issues during client onboarding by 60%.
- Built React and TypeScript interfaces for the import APIs, implemented typed client-side validation and reusable UI components, and added automated coverage reporting to the team's CI pipeline.

Software Developer | Cognizant Technology Solutions (Client: AIG UK) | Hyderabad, India

Feb 2021 to Jul 2022

- Developed features across AIG UK's commercial policy lifecycle platform using Java, Spring Boot, and Hibernate/JPA on SQL Server, supporting 10+ core policy workflows with over 90% JUnit-driven test coverage.
- Designed Spring Boot REST API integrations with third-party insurance systems and built customer-facing portals using Angular, React, TypeScript, HTML5, Bootstrap, and CSS3.
- Built Kafka producer and consumer services for asynchronous workflows between policy lifecycle microservices, decoupling previously blocking dependencies and improving fault tolerance.
- Performed query tuning, applied caching strategies, and optimized TypeScript frontend bundles to reduce page load times by 30% while managing CI/CD pipelines with Azure DevOps.

EDUCATION

Master of Science in Computer Science

Aug 2022 to May 2024

University of Oklahoma, Norman, OK • GPA: 3.7/4.0

Bachelor of Technology in Computer Science & Engineering

Jul 2017 to Jun 2021

Vardhaman College of Engineering, Hyderabad, India • GPA: 8.12/10

PROJECTS

AI-Augmented Web Crawler | Python, AsyncIO, LangChain, OpenAI APIs, Redis, BeautifulSoup

- Designed and built a high-concurrency web crawler using AsyncIO and thread-pool fetching, leveraging LLM content classification and structured data extraction to replace brittle regex/CSS selector logic and improve resilience to target site layout changes.
- Implemented Redis-backed deduplication, task queues, and checkpointing for fault-tolerant parallel runs.

Sensitive Data Redaction Tool | Python, spaCy, scikit-learn, Regex, SQLite

- Developed a backend tool to identify and redact sensitive PII in documents using a Named Entity Recognition pipeline built with spaCy and scikit-learn combined with regex pattern matching, with SQLite for state management and audit logging.

PUBLICATIONS

- Image Dehazing for Surveillance Systems, 4th Smart Cities Symposium (SCS 2021), IET Digital Library [\[Link\]](#)
- URL Shortening Service Using Flask & Base62 Algorithm, International Conference on Intelligent Computing and Networking (ICICN 2021).